

RESEARCH LETTER

ESAM Loss of Function and Congenital Neurovascular Injury: Strengthening the Case for a Recognizable Clinical Phenotype

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Received: 21 November 2025 | **Revised:** 3 January 2026 | **Accepted:** 6 January 2026

Neurodevelopmental disorders (NDDs) are a major unmet challenge in child health, affecting millions and often causing lifelong disability. Despite their clinical impact, the molecular mechanisms behind many NDDs remain poorly defined, limiting diagnostic and therapeutic progress. Emerging evidence highlights the blood–brain barrier (BBB), particularly endothelial junction integrity as a crucial factor in early brain vulnerability. Disruption of these junctions has been linked to both acute perinatal injury and chronic neurodevelopmental syndromes [1]. Recent mechanistic studies highlight *Endothelial Cell Selective Adhesion Molecule (ESAM)*'s essential role in endothelial adhesion, angiogenesis, and BBB integrity, with loss of *ESAM* disrupting junctional stability, increasing vascular permeability and hemorrhage risk, and impairing fetal cerebral microvascular. By presenting three individuals from two unrelated Turkish families, this letter contributes to the expanding recognition of *ESAM* as a gene crucial for BBB stability and fetal cerebral vascular development.

Case 1 and Case 3, siblings born to consanguineous parents and are homozygous for a splice-site variant (NM_138961.3:c.451+5G>C), exhibited severe developmental impairment beginning in infancy, early refractory seizures (at 6 and 4 months of age), and a combination of porencephaly, ventriculomegaly, corpus callosum hypoplasia, and intracranial calcifications (Figure 1). This variant was classified as a Variant of Uncertain Significance (VUS) based on ACMG criteria (PM2 [Moderate], PP3 [Supporting]). Their shared dysmorphic features including bitemporal narrowing, elongated eyelashes, and

a bulbous nasal tip further reinforce the presence of a recognizable phenotype. Tragically, the older sibling succumbed to complications of recurrent severe pneumonia at age 11, underscoring the multisystem fragility accompanying profound neurodisability.

Case 2, who was born to non-consanguineous parents, carried a novel homozygous nonsense variant (NM_138961.3:c.605T>G; p.Leu202Ter), classified as Pathogenic based on ACMG criteria (PVS1 [Very Strong], PM2 [Moderate]). She displayed a nearly identical clinical trajectory (at 6 months of age), beginning with prenatally detected hydrocephalus and early NICU admission for respiratory distress (at 3 months of age) (Figure 1). Her daily focal seizures, bilateral cataracts, microcephaly, and highly overlapping neuroimaging findings point to a shared pathogenic mechanism rooted in compromised vascular–neurodevelopmental pathways.

We evaluated the functional effects of the *ESAM* variants using an Ensembl VEP, SpliceAI, and Pangolin. The splice-region variant c.451+5G>C showed a high predicted probability of disrupting the canonical donor site (SpliceAI donor-loss Δ =0.99; Pangolin splice-loss=0.80), with predicted cryptic donor activation. These models suggest exon 3 skipping or alternative splice-site usage, likely resulting in a frameshift and nonsense-mediated decay. The p.Leu202Ter variant truncates the protein within the second Ig-like domain and is predicted to result in loss of function. Both variants were absent in gnomAD and had high CADD scores, supporting pathogenicity.

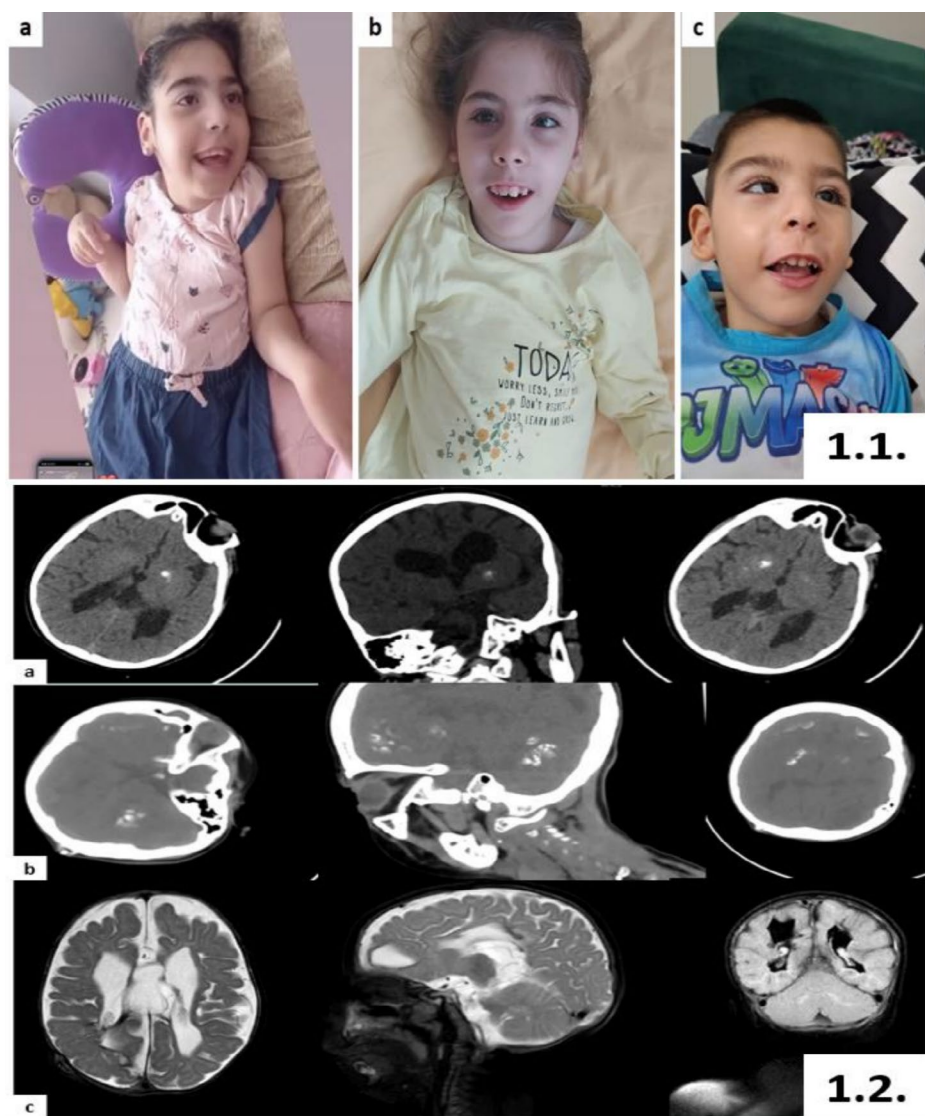


FIGURE 1 | (1) Facial features of the patients. (a) 11-year-old Kurdish female (Case 1). (b) 10-year-old Turkish female (Case 2). (c) 2-year-old male, younger sibling of Case 1 (Case 3). (2) Neuroimaging findings of three affected cases. (a) Case 1 (at age 2): Brain CT images demonstrating intracranial calcifications, ventriculomegaly, and periventricular white matter changes. (b) Case 2 (at age 9): CT scans showing extensive bilateral intracranial calcifications and cortical atrophy. (c) Case 3 (at age 7): Axial, sagittal, and coronal T2-weighted MRI images revealing ventriculomegaly, periventricular leukomalacia, and thinning of the corpus callosum. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/cge.70140)]

Prenatal or perinatal intracranial hemorrhage is a consistent hallmark of *ESAM* deficiency, reflecting its role in fetal cerebral vascular stability and producing a severe yet often childhood-survivable phenotype distinct from the more fulminant *JAM3*- or *OCLN*-related disorders. Together, these reports including the present Turkish families expand the *ESAM* mutational spectrum and support the consensus that complete *ESAM* loss-of-function causes a severe congenital-onset neurovascular disorder, consistent with mechanistic evidence of absent endothelial *ESAM* and impaired tubulogenesis [2] and recurrent clinical associations with perinatal stroke and congenital intracranial hemorrhage, particularly in consanguineous populations [3, 4].

Across the cases, a characteristic neuroimaging pattern (por-encephaly, periventricular leukomalacia, ventriculomegaly, intracranial calcifications, and corpus callosum hypoplasia (Figure 1)) emerges as a strong diagnostic clue for *ESAM*-related

disease, reflecting both acute hemorrhage and chronic vascular insufficiency and warranting *ESAM* inclusion in genomic testing when this constellation is identified.

Based on Lecca et al. and current evidence, *ESAM* deficiency-related neurodevelopmental disease likely arises from the combined effects of perinatal intracranial hemorrhage, impaired angiogenesis, and chronic BBB instability [2]. Perinatal hemorrhage likely represents the initiating insult with direct tissue damage and inflammation, while defective endothelial tube formation and migration during critical developmental windows impair normal vascular maturation, and persistent BBB dysfunction sustains neuroinflammation and oxidative stress.

In conclusion, the three individuals summarized in this manuscript reinforce *ESAM* deficiency as a severe but distinct member of the tight junctionopathy spectrum. As genomic diagnostics

become further integrated into neonatal and pediatric care, *ESAM* sequencing should be considered a critical component of the evaluation of unexplained perinatal stroke, porencephaly, or congenital neurodevelopmental syndromes.

Funding

The authors have nothing to report.

Ethics Statement

This study was conducted in Kartal Dr. Lutfi Kırdar Training and Research Hospital and the ethical approval was waived by the policies of the hospital.

Consent

Informed consent was obtained from the participants to participate in the current study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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